



Module 3: Protocols and Models

Telecom Academy
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Introduction to Networks v7.0
(ITN)



Module Objectives

Module Title: Protocols and Models

Module Objective: Explain how network protocols enable devices to access local and remote network resources.

Topic Title	Topic Objective
The Rules	Describe the types of rules that are necessary to successfully communicate.
Protocols	Explain why protocols are necessary in network communication.
Protocol Suites	Explain the purpose of adhering to a protocol suite.
Standards Organizations	Explain the role of standards organizations in establishing protocols for network interoperability.
Reference Models	Explain how the TCP/IP model and the OSI model are used to facilitate standardization in the communication process.
Data Encapsulation	Explain how data encapsulation allows data to be transported across the network.
Data Access	Explain how local hosts access local resources on a network.

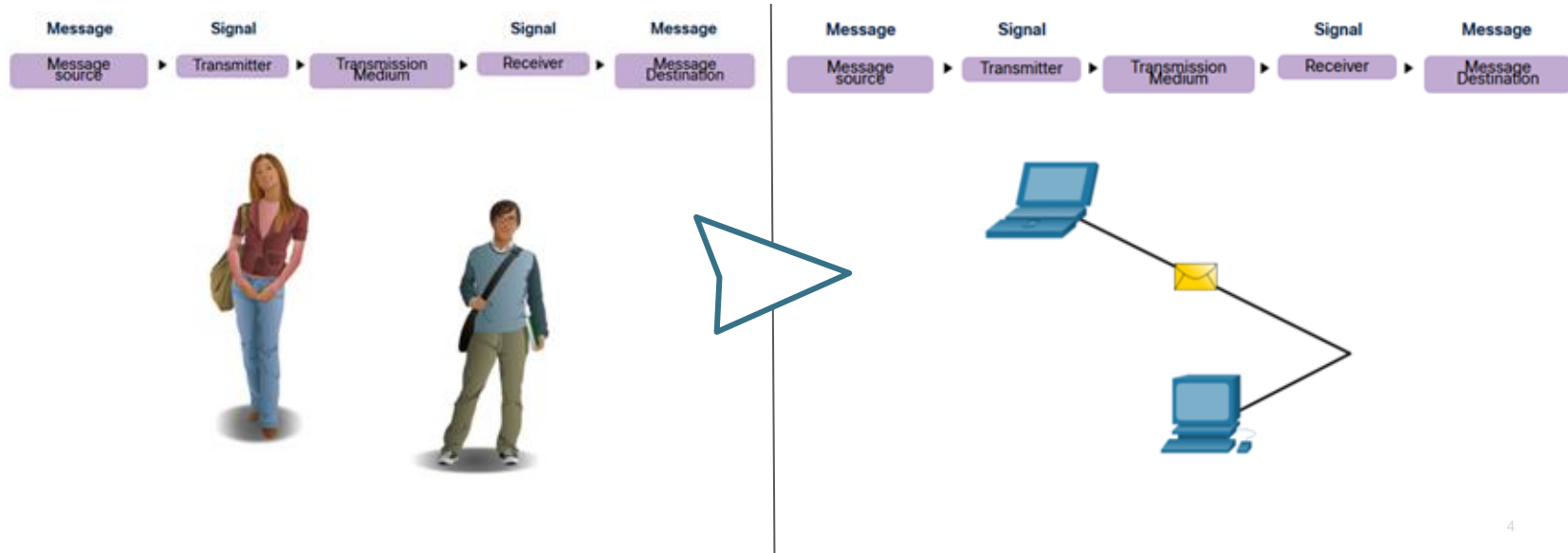


3.1 The Rules

The Rules

Communications Protocols

- All communications are governed by protocols.
- Protocols are the rules that communications will follow.
- These rules will vary depending on the protocol.



The Rules

Rule Establishment

- Individuals must use established rules or agreements to govern the conversation.
- The first message is difficult to read because it is not formatted properly. The second shows the message properly formatted

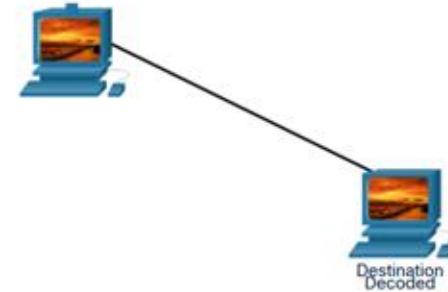
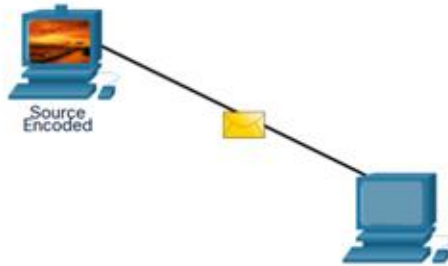
humans communication between govern rules. It is verydifficult tounderstand messages that are not correctly formatted and donot follow the established rules and protocols. A estrutura da gramatica, da lingua, da pontuacao e do sentence faz a configuracao humana compreensivel por muitos individuos diferentes.

Rules govern communication between humans. It is very difficult to understand messages that are not correctly formatted and do not follow the established rules and protocols. The structure of the grammar, the language, the punctuation and the sentence make the configuration humanly understandable for many different individuals.

The Rules

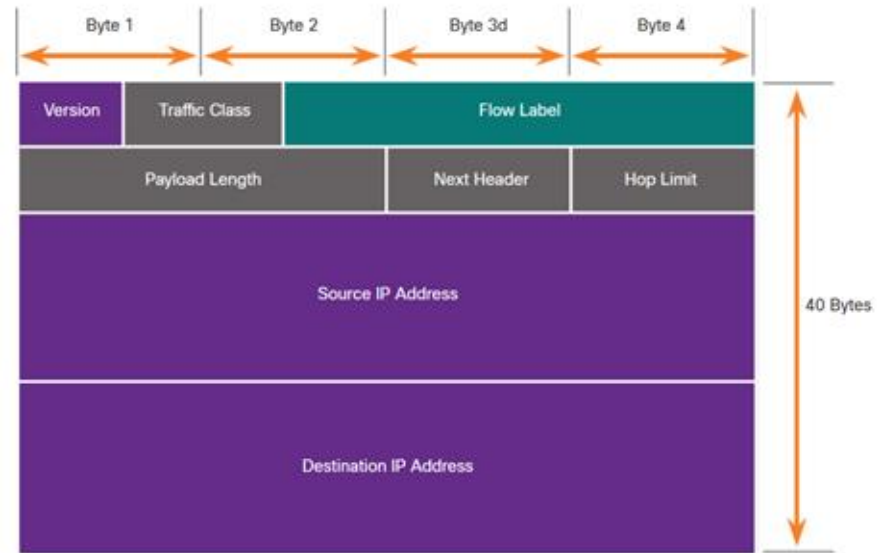
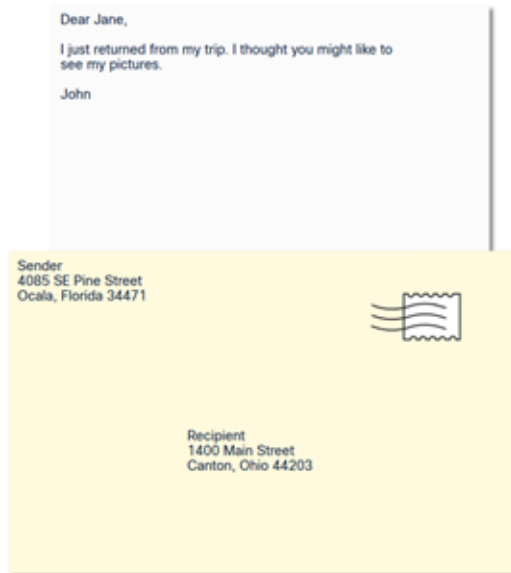
Message Encoding

- Encoding is the process of converting information into another acceptable form for transmission.
- Decoding reverses this process to interpret the information.



Message Formatting and Encapsulation

- When a message is sent, it must use a specific format or structure.
- Message formats depend on the type of message and the channel that is used to deliver the message.

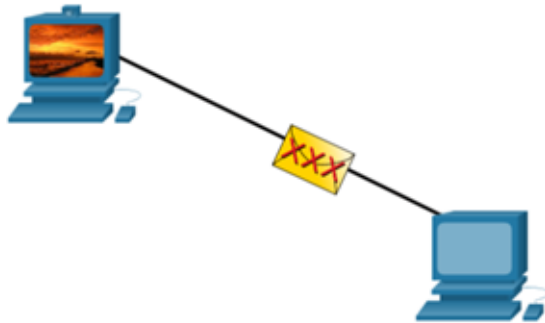


The Rules

Message Size

Encoding between hosts must be in an appropriate format for the medium.

- Messages sent across the network are converted to bits
- The bits are encoded into a pattern of light, sound, or electrical impulses.
- The destination host must decode the signals to interpret the message.



Message Timing

Message timing includes the following:

Flow Control – Manages the rate of data transmission and defines how much information can be sent and the speed at which it can be delivered.

Response Timeout – Manages how long a device waits when it does not hear a reply from the destination.

Access method - Determines when someone can send a message.

- There may be various rules governing issues like “collisions”. This is when more than one device sends traffic at the same time and the messages become corrupt.
- Some protocols are proactive and attempt to prevent collisions; other protocols are reactive and establish a recovery method after the collision occurs.

Message Delivery Options

Message delivery may use one of the following methods:

- **Unicast** – one to one communication
- **Multicast** – one to many, typically not all
- **Broadcast** – one to all

Note: Broadcasts are used in IPv4 networks, but are not an option for IPv6. Later we will also see “Anycast” as an additional delivery option for IPv6.



Unicast

Multicast

Broadcast



Unicast

Multicast

Broadcast



Unicast

Multicast

Broadcast

3.2 Protocols

Network Protocol Overview

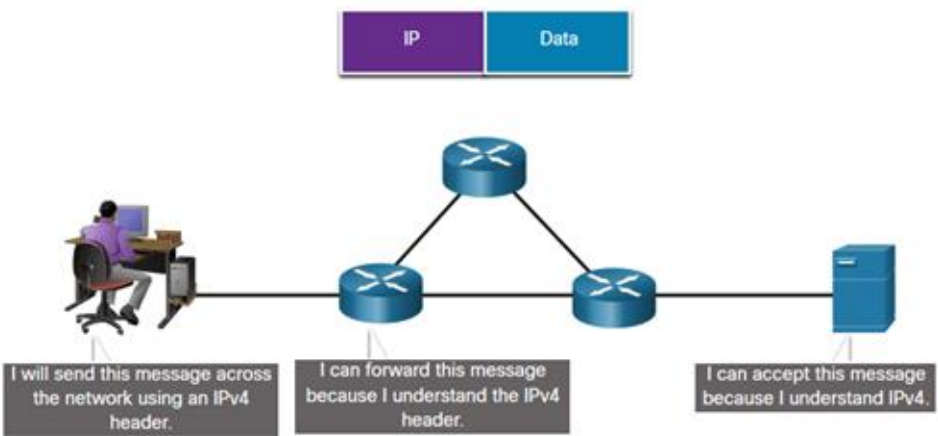
Network protocols define a common set of rules.

- Can be implemented on devices in:
 - Software
 - Hardware
 - Both
- Protocols have their own:
 - Function
 - Format
 - Rules

Protocol Type	Description
Network Communications	enable two or more devices to communicate over one or more networks
Network Security	secure data to provide authentication, data integrity, and data encryption
Routing	enable routers to exchange route information, compare path information, and select best path
Service Discovery	used for the automatic detection of devices or services

Network Protocol Functions

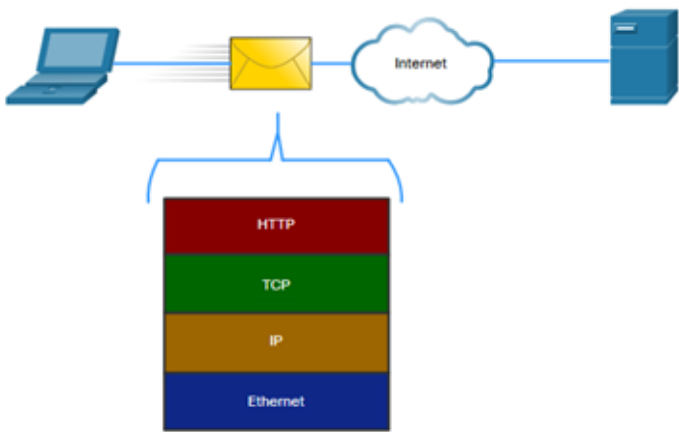
- Devices use agreed-upon protocols to communicate .
- Protocols may have may have one or functions.



Function	Description
Addressing	Identifies sender and receiver
Reliability	Provides guaranteed delivery
Flow Control	Ensures data flows at an efficient rate
Sequencing	Uniquely labels each transmitted segment of data
Error Detection	Determines if data became corrupted during transmission
Application Interface	Process-to-process communications between network applications

Protocol Interaction

- Networks require the use of several protocols.
- Each protocol has its own function and format.



Protocol	Function
Hypertext Transfer Protocol (HTTP)	▪ Governs the way a web server and a web client interact ▪ Defines content and format
Transmission Control Protocol (TCP)	▪ Manages the individual conversations ▪ Provides guaranteed delivery ▪ Manages flow control
Internet Protocol (IP)	Delivers messages globally from the sender to the receiver
Ethernet	Delivers messages from one NIC to another NIC on the same Ethernet Local Area Network (LAN)

3.3 Protocol Suites

Protocol Suites

Evolution of Protocol Suites

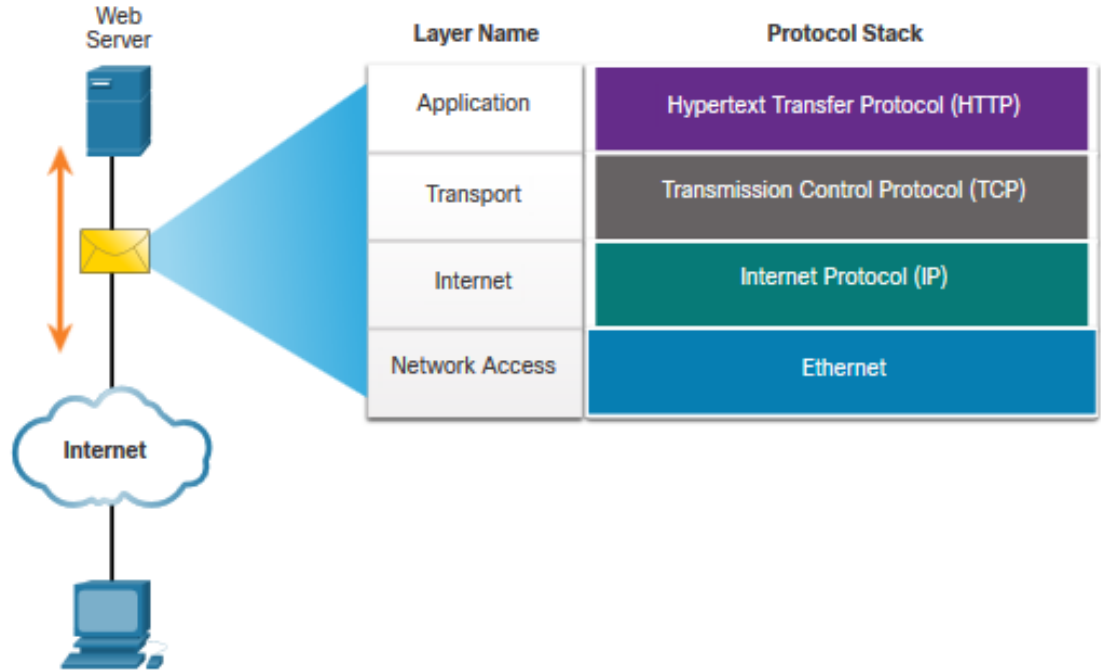
There are several protocol suites.

- **Internet Protocol Suite or TCP/IP**- The most common protocol suite and maintained by the Internet Engineering Task Force (IETF)
- **Open Systems Interconnection (OSI) protocols**- Developed by the International Organization for Standardization (ISO) and the International Telecommunications Union (ITU)
- **AppleTalk**- Proprietary suite release by Apple Inc.
- **Novell NetWare**- Proprietary suite developed by Novell Inc.

TCP/IP Layer Name	TCP/IP	ISO	AppleTalk	Novell Netware
Application	HTTP DNS DHCP FTP	ACSE ROSE TRSE SESE	AFP	NDS
Transport	TCP UDP	TP0 TP1 TP2 TP3 TP4	ATP AEP NBP RTMP	SPX
Internet	IPv4 IPv6 ICMPv4 ICMPv6	CONP/CMNS CLNP/CLNS	AARP	IPX
Network Access	Ethernet ARP WLAN			

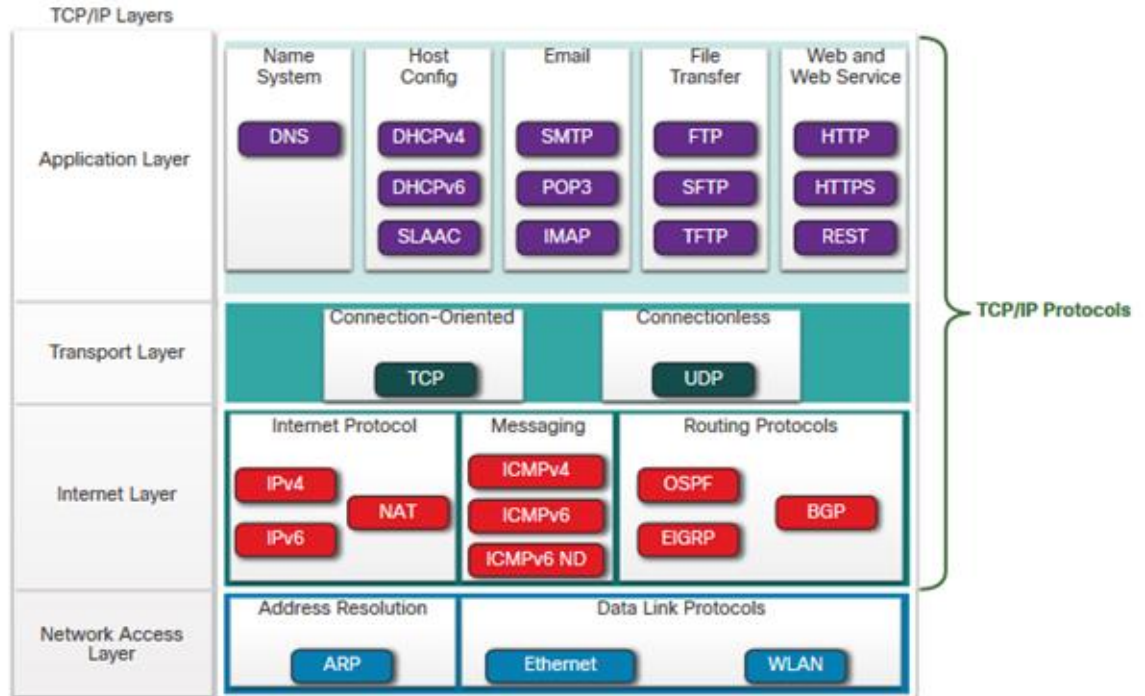
TCP/IP Protocol Example

- TCP/IP protocols operate at the application, transport, and internet layers.
- The most common network access layer LAN protocols are Ethernet and WLAN (wireless LAN).



TCP/IP Protocol Suite

- TCP/IP is the protocol suite used by the internet and includes many protocols.
- TCP/IP is:
 - An open standard protocol suite that is freely available to the public and can be used by any vendor
 - A standards-based protocol suite that is endorsed by the networking industry and approved by a standards organization to ensure interoperability



3.4 Standards Organizations

Standards Organizations

Open Standards



I E T F



Open standards encourage:

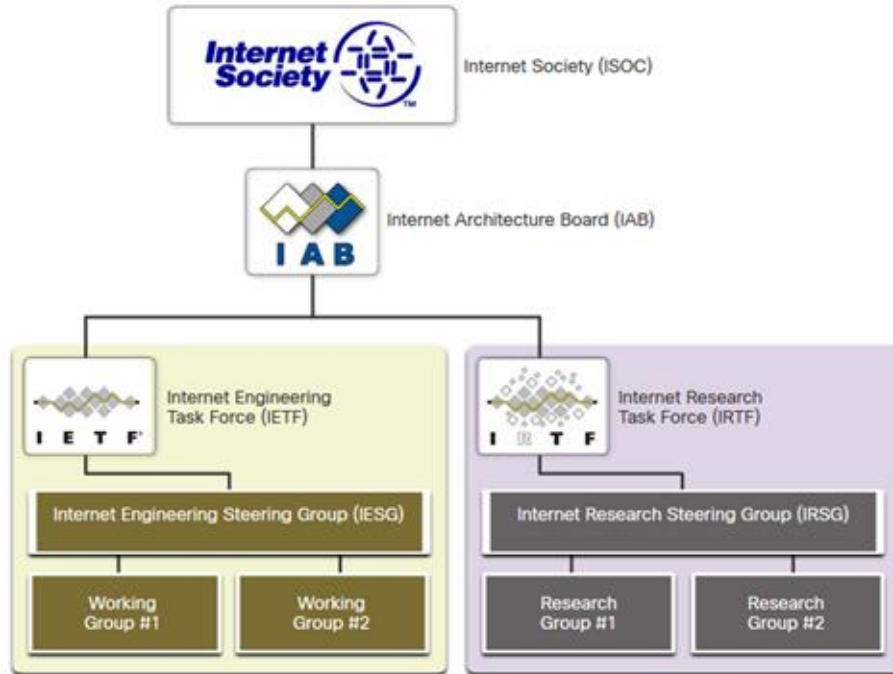
- interoperability
- competition
- innovation

Standards organizations are:

- vendor-neutral
- non-profit organizations
- established to develop and promote the concept of open standards.

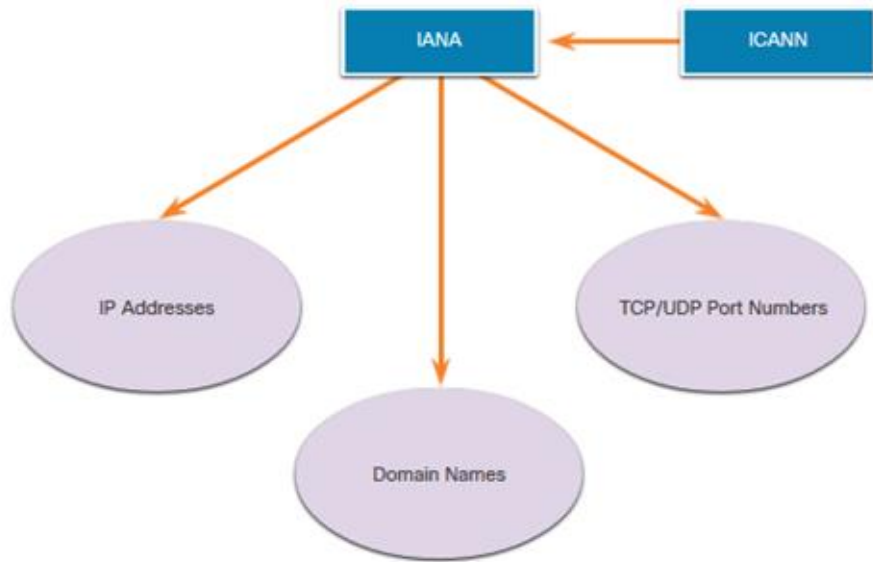
Standards Organizations

Internet Standards



- **Internet Society (ISOC)** - Promotes the open development and evolution of internet
- **Internet Architecture Board (IAB)** - Responsible for management and development of internet standards
- **Internet Engineering Task Force (IETF)** - Develops, updates, and maintains internet and TCP/IP technologies
- **Internet Research Task Force (IRTF)** - Focused on long-term research related to internet and TCP/IP protocols

Internet Standards (Cont.)



Standards organizations involved with the development and support of TCP/IP

- **Internet Corporation for Assigned Names and Numbers (ICANN)** - Coordinates IP address allocation, the management of domain names, and assignment of other information
- **Internet Assigned Numbers Authority (IANA)** - Oversees and manages IP address allocation, domain name management, and protocol identifiers for ICANN

IEEE 802.3

21 languages

Article Talk

Read Edit View history Tools

From Wikipedia, the free encyclopedia

This article is about the standards working group. For Ethernet frame data format, see [Ethernet frame](#).

IEEE 802.3 is a [working group](#) and a collection of standards defining the [physical layer](#) and [data link layer](#)'s [media access control](#) (MAC) of wired [Ethernet](#). The standards are produced by the working group of [Institute of Electrical and Electronics Engineers](#) (IEEE). This is generally a [local area network](#) (LAN) technology with some [wide area network](#) (WAN) applications. Physical connections are made between nodes and/or infrastructure devices ([hubs](#), [switches](#), [routers](#)) by various types of copper or [fiber cable](#).

802.3 is a technology that supports the [IEEE 802.1](#) network architecture.

802.3 also defines LAN access method using [CSMA/CD](#).

Communication standards

[\[edit \]](#)

Ethernet standard	Date	Description
Experimental Ethernet	1973 ^[1]	2.94 Mbit/s (367 kB/s) over coaxial cable (coax) bus. Single byte node address unique only to individual network.
Ethernet I (DIX v1.0)	1980	10 Mbit/s (1.25 MB/s) over thick coax. Frames have a Type field. This frame format is used on all forms of Ethernet by protocols in the Internet protocol suite . Six byte MAC address.
Ethernet II (DIX v2.0)	1982	
IEEE 802.3 standard	1983	10BASE5 10 Mbit/s (1.25 MB/s) over thick coax. Same as Ethernet II (above) except Type field is replaced by Length, and an 802.2 LLC header follows the 802.3 header. Based on the CSMA/CD Process.
802.3a	1985	10BASE2 10 Mbit/s (1.25 MB/s) over thin Coax (a.k.a. thinnet or cheapernet)
802.3b	1985	10BROAD36
802.3c	1985	10 Mbit/s (1.25 MB/s) repeater specs
802.3-1985	1985	a revision of the base standard from 1983
802.3d	1987	Fiber-optic inter-repeater link
802.3e	1987	1BASE5 or StarLAN , first use of (voice-grade) twisted pair cabling. 1 Mbit/s, maximum reach of 250 to 500 m
802.3i	1990	10BASE-T 10 Mbit/s (1.25 MB/s) over twisted pair
802.3j	1993	10BASE-F 10 Mbit/s (1.25 MB/s) over optical fiber
802.3q	1993	GDMO (ISO/IEC 10165-4) format for Layer Managed Objects
802.3u	1995	100BASE-TX , 100BASE-T4 , 100BASE-FX Fast Ethernet at 100 Mbit/s (12.5 MB/s) with autonegotiation
802.3x	1997	Full Duplex and flow control ; also incorporates DIX framing, so there's no longer a DIX/802.3 split
802.3y	1998	100BASE-T2 100 Mbit/s (12.5 MB/s) over voice-grade twisted pair
802.3z	1998-07	1000BASE-X Gbit/s Ethernet over optical fiber at 1 Gbit/s (125 MB/s)
802.3-1998	1998-07	(802.3aa) A revision of base standard incorporating the above amendments and errata

Internet Protocol version 4

47 languages

Article Talk

Read Edit View history Tools

From Wikipedia, the free encyclopedia

Internet Protocol version 4 (IPv4) is the fourth version of the *Internet Protocol* (IP). It is one of the core protocols of standards-based *internetworking* methods in the *Internet* and other *packet-switched* networks. IPv4 was the first version deployed for production on *SATNET* in 1982 and on the *ARPANET* in January 1983. It is still used to route most *Internet traffic* today,^[1] even with the ongoing deployment of *Internet Protocol version 6* (IPv6),^[2] its successor.

IPv4 uses a **32-bit** address space which provides 4,294,967,296 (2^{32}) unique addresses, but large blocks are reserved for special networking purposes.^{[3][4]}

History

Internet Protocol version 4 is described in *IETF* publication RFC 791 (September 1981), replacing an earlier definition of January 1980 (RFC 760). In March 1982, the US Department of Defense decided on the *Internet Protocol Suite* (TCP/IP) as the standard for all military *computer networking*.^[5]

Purpose

The Internet Protocol is the protocol that defines and enables *internetworking* at the *Internet layer* of the Internet Protocol Suite. In essence it forms the Internet. It uses a logical addressing system and performs *routing*, which is the forwarding of packets from a source host to the next router that is one hop closer to the intended destination host on another network.

IPv4 is a *connectionless* protocol, and operates on a *best-effort delivery* model, in that it does not guarantee delivery, nor does it assure proper sequencing or avoidance of duplicate delivery. These aspects, including data integrity, are addressed by an *upper layer* transport protocol, such as the *Transmission Control Protocol* (TCP).

Addressing

For broader coverage of this topic, see *IP address*.

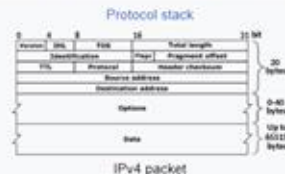
IPv4 uses 32-bit addresses which limits the *address space* to 4 294 967 296 (2^{32}) addresses.

IPv4 reserves special address blocks for *private networks* (~18 million addresses) and *multicast* addresses (~270 million addresses).

Address representations

IPv4 addresses may be represented in any notation expressing a 32-bit integer value. They are most often written in *decimal notation*, which consists of four *octets* of the address arranged

Internet Protocol version 4



Abbreviation IPv4

Purpose internetworking protocol

Developer(s) DARPA

Introduction 1981; 42 years ago

Influenced IPv6

OSI layer Network layer

RFC(s) 791 et

Internet protocol suite

Application layer

BGP · DHCP (v6) · DNS · FTP · HTTP (HTTP/3) · HTTPS · IMAP · IRC · LDAP · MGCP · MQTT · NNTP · NTP · OSPF · POP · PTP · QNC/RPC · RTP · RTSP · SIP · SMTP · SNMP · SSH · Telnet · TLS/SSL · XMPP · more...

Transport layer

TCP · UDP · DCCP · SCTP · RSVP · QUIC · more...

Internet layer

IP (v4 · v6) · ICMP (v6) · NDP · ECN · IGMP · IPsec · more...

Link layer

ARP · Tunnels · PPP · MAC · more...

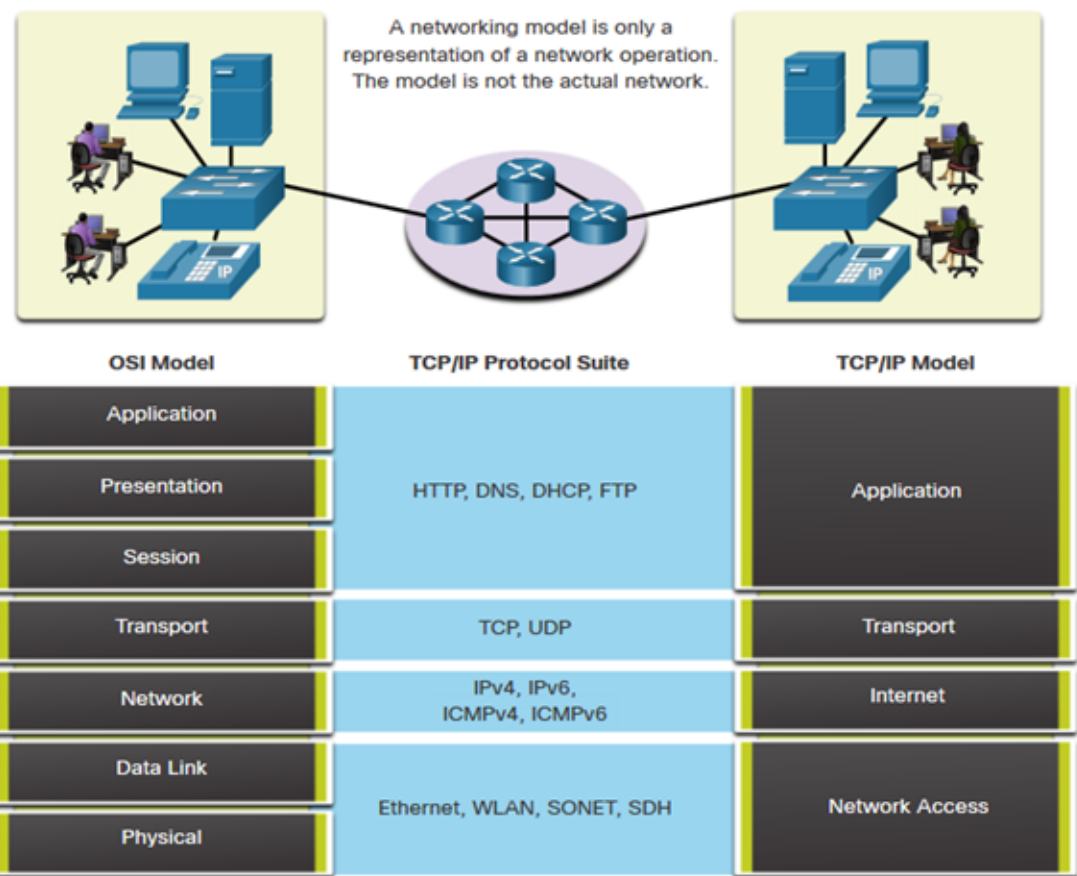
V · T · E



Most significant bit	172	16	254	1	Dot-decimal notation
	↓	↓	↓	↓	Binary
	10101100	00010000	11111110	00000001	

3.5 Reference Models

The Benefits of Using a Layered Model



Complex concepts such as how a network operates can be difficult to explain and understand. For this reason, a layered model is used.

Two layered models describe network operations:

- Open System Interconnection (OSI) Reference Model
- TCP/IP (Reference Model)

The OSI Reference Model

OSI Model Layer	Description
7 - Application	Contains protocols used for process-to-process communications.
6 - Presentation	Provides for common representation of the data transferred between application layer services.
5 - Session	Provides services to the presentation layer and to manage data exchange.
4 - Transport	Defines services to segment, transfer, and reassemble the data for individual communications.
3 - Network	Provides services to exchange the individual pieces of data over the network.
2 - Data Link	Describes methods for exchanging data frames over a common media.
1 - Physical	Describes the means to activate, maintain, and de-activate physical connections.



OSI-Model

Open Systems Interconnection model

The OSI Model was designed in 1977 and is part of the ISO specifications since 1984.
Each OSI layer serves a unique task for network communication.

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Layer 7  **Application Layer**

Layer 6  **Presentation Layer**

Layer 5  **Session Layer**

Layer 4  **Transport Layer**

Layer 3  **Network Layer**

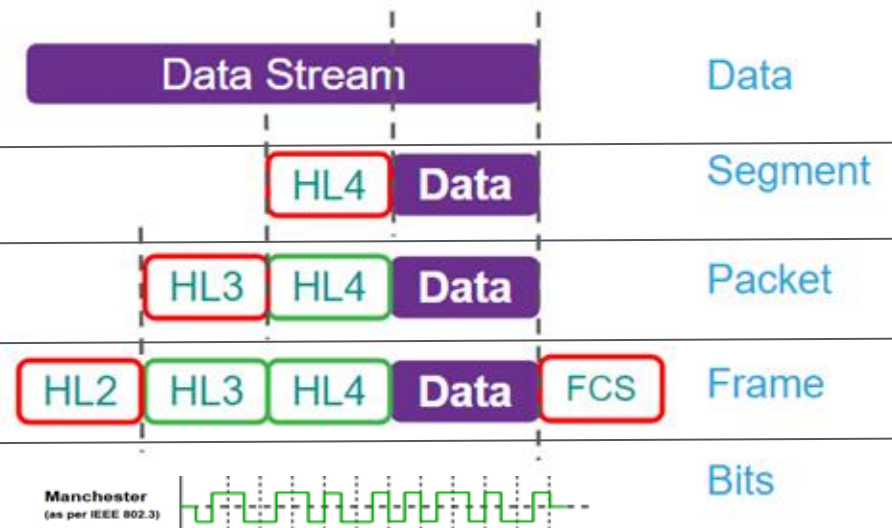
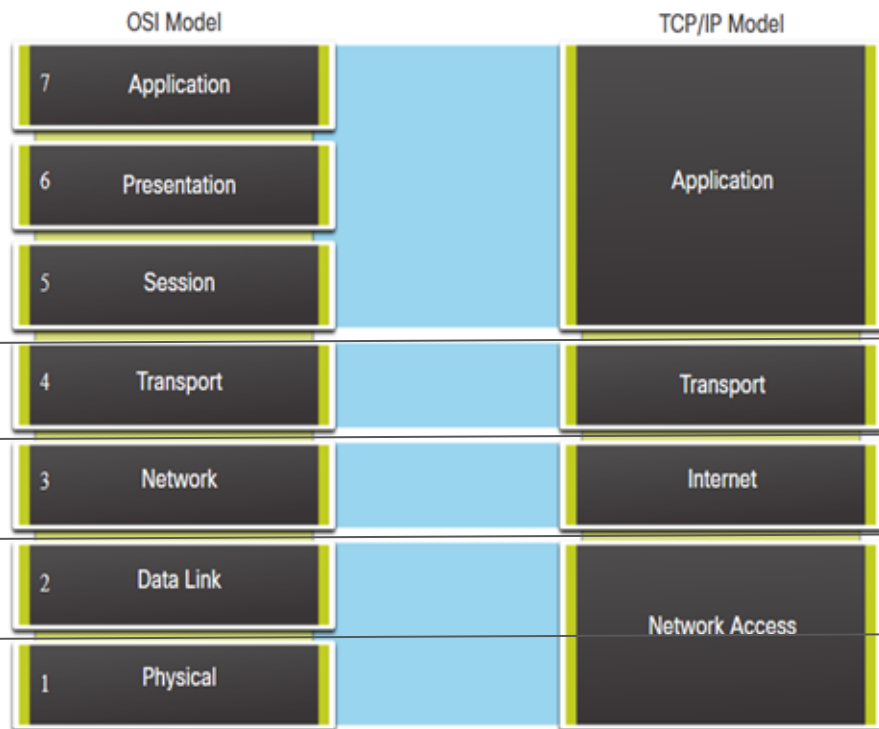
Layer 2  **Data Link Layer**

Layer 1  **Physical Layer**

The TCP/IP Reference Model

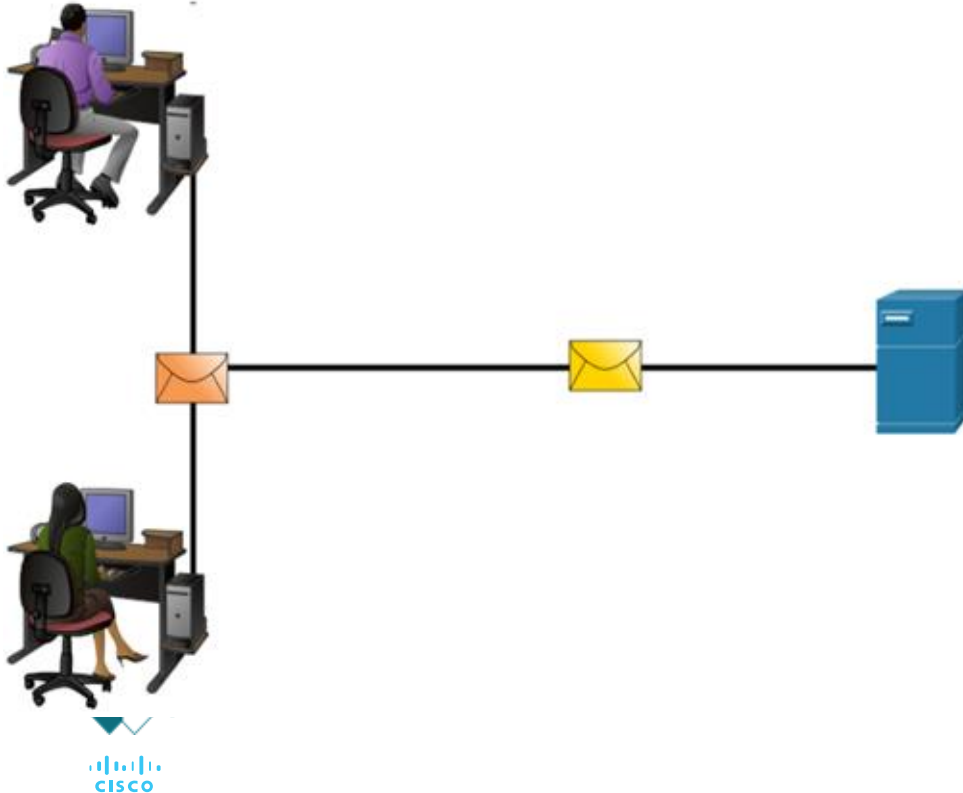
TCP/IP Model Layer	Description
Application	Represents data to the user, plus encoding and dialog control.
Transport	Supports communication between various devices across diverse networks.
Internet	Determines the best path through the network.
Network Access	Controls the hardware devices and media that make up the network.

OSI and TCP/IP Model Comparison



3.6 Data Encapsulation

Segmenting Messages



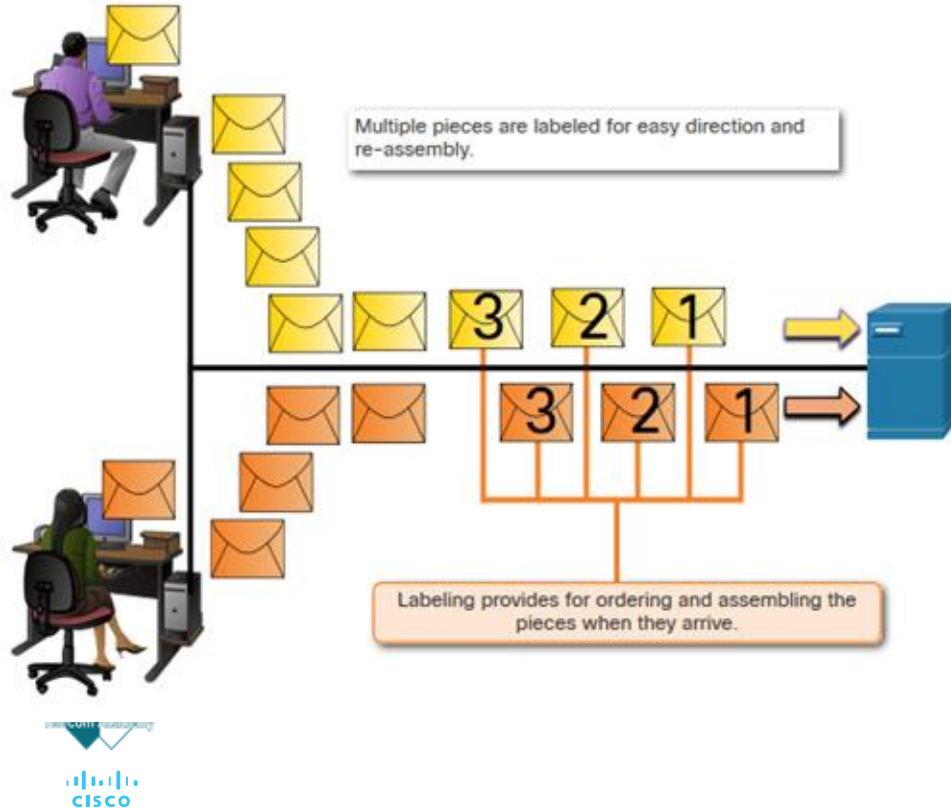
Segmenting is the process of breaking up messages into smaller units. Multiplexing is the processes of taking multiple streams of segmented data and interleaving them together.

Segmenting messages has two primary benefits:

- **Increases speed** - Large amounts of data can be sent over the network without tying up a communications link.
- **Increases efficiency** - Only segments which fail to reach the destination need to be retransmitted, not the entire data stream.

Data Encapsulation

Sequencing



Sequencing messages is the process of numbering the segments so that the message may be reassembled at the destination.

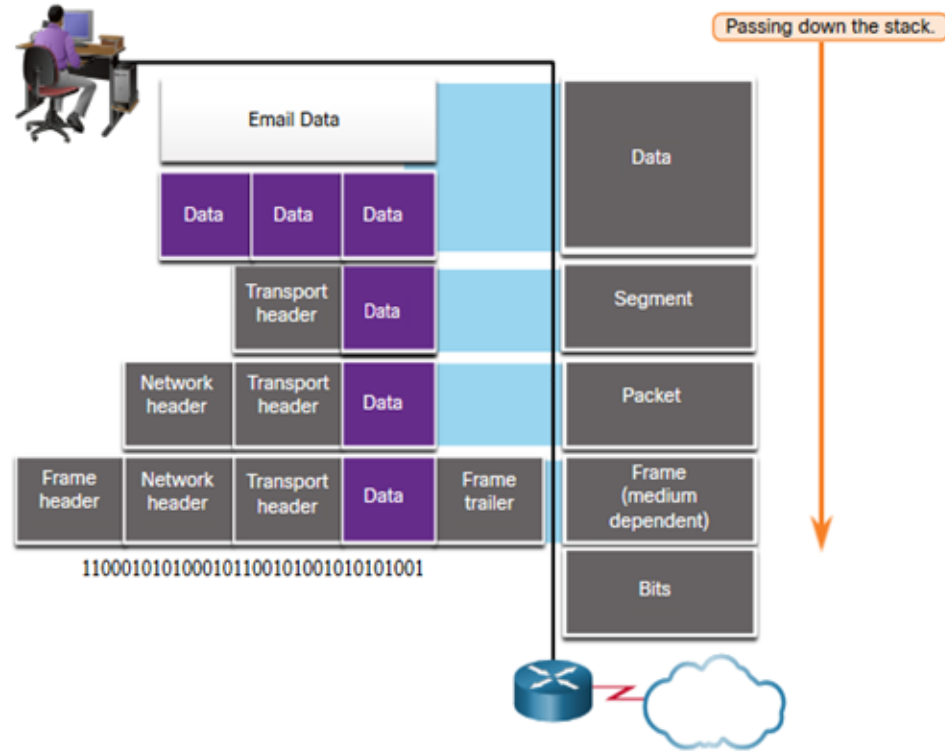
TCP is responsible for sequencing the individual segments.

Data Encapsulation

Protocol Data Units

Encapsulation is the process where protocols add their information to the data.

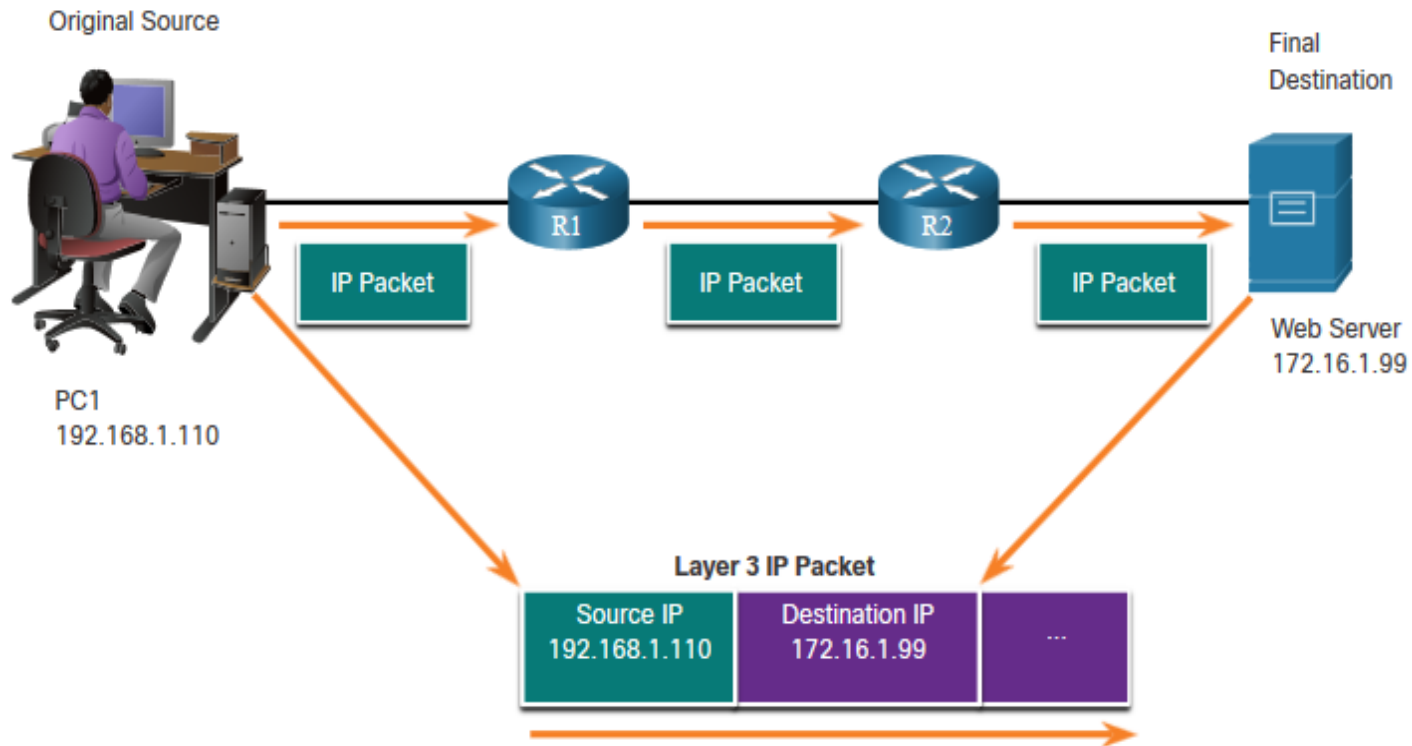
- At each stage of the process, a PDU has a different name to reflect its new functions.
- There is no universal naming convention for PDUs, in this course, the PDUs are named according to the protocols of the TCP/IP suite.
- PDUs passing down the stack are as follows:
 1. Data (Data Stream)
 2. Segment
 3. Packet
 4. Frame
 5. Bits (Bit Stream)



3.7 Data Access

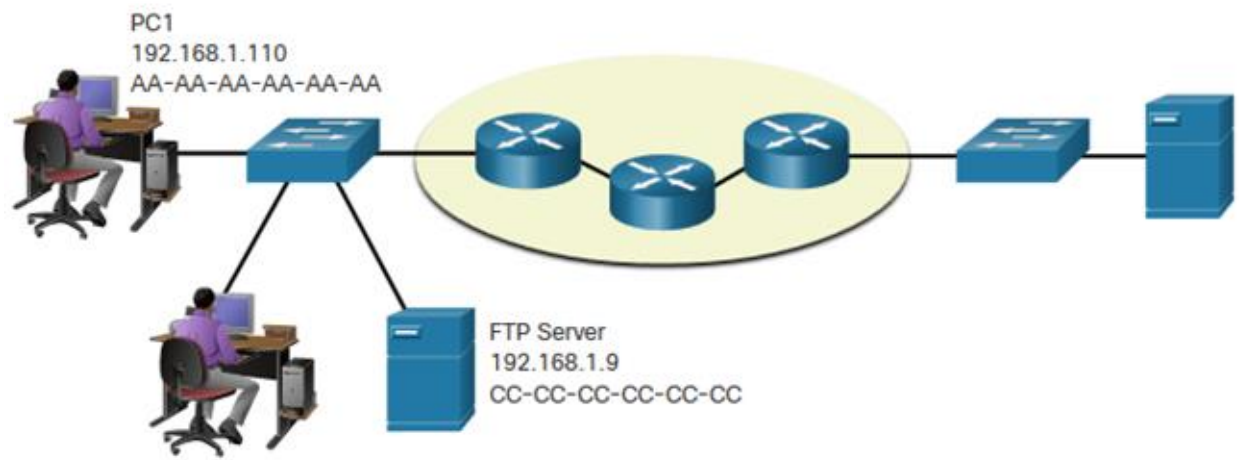
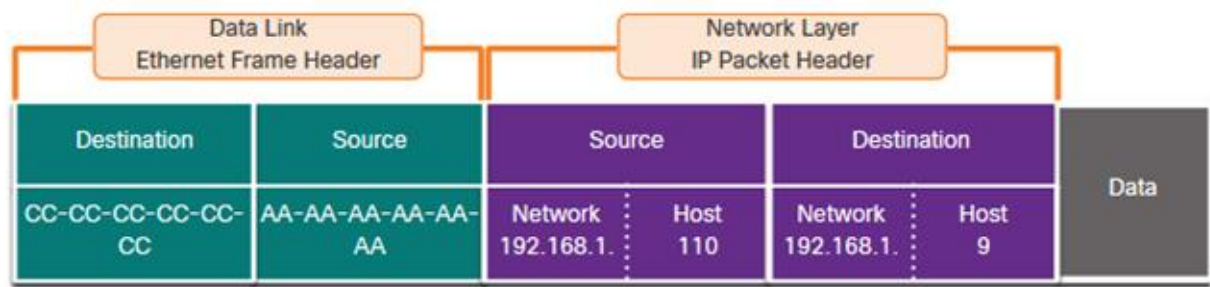
Data Access

Layer 3 Logical Address

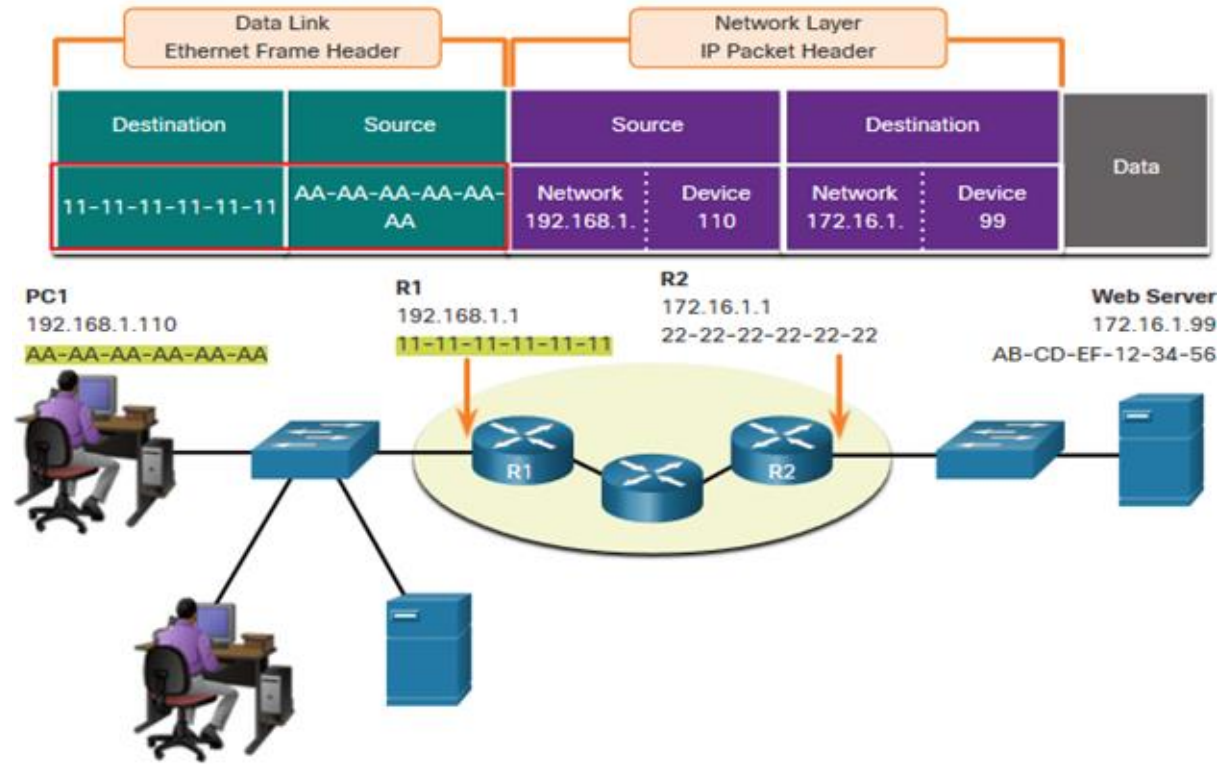


Data Access

Devices on the Same Network

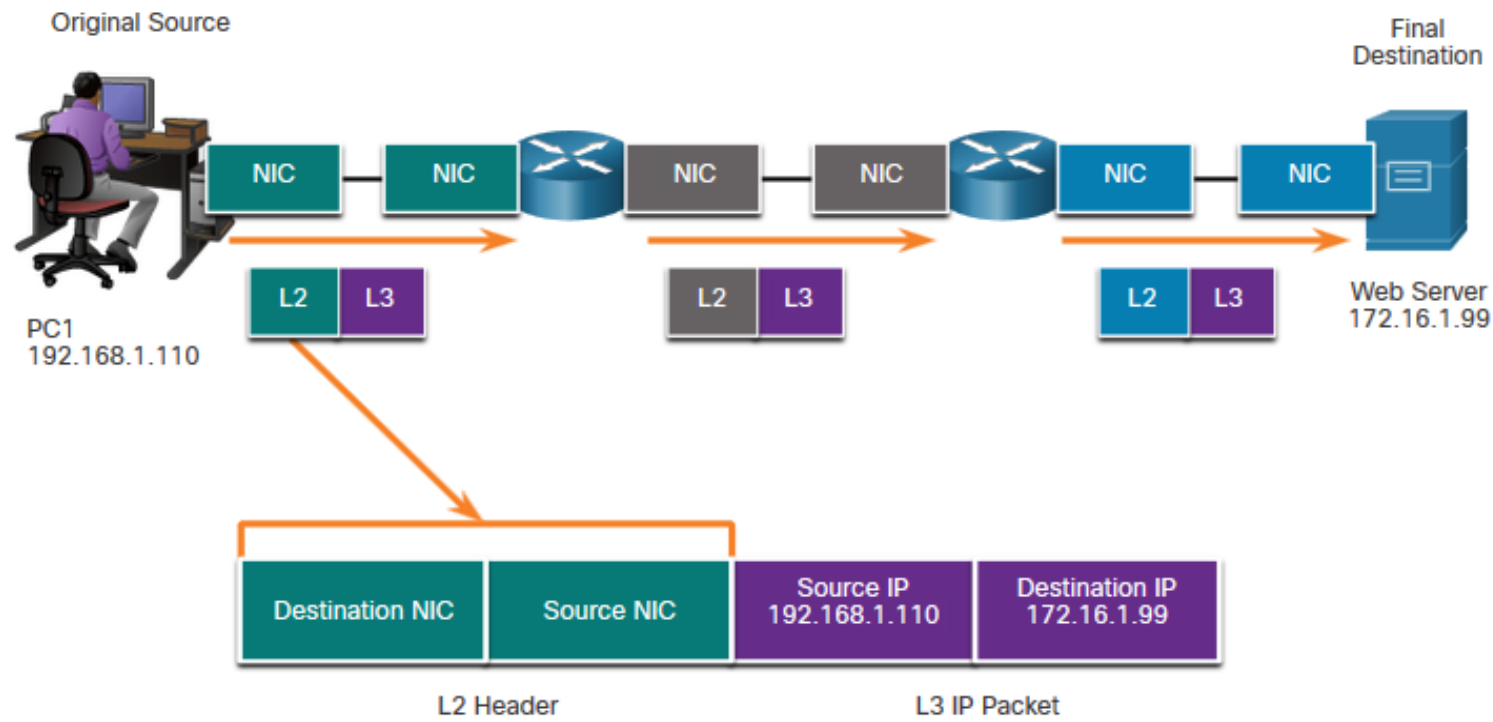


Role of the Data Link Layer Addresses: Different IP Networks



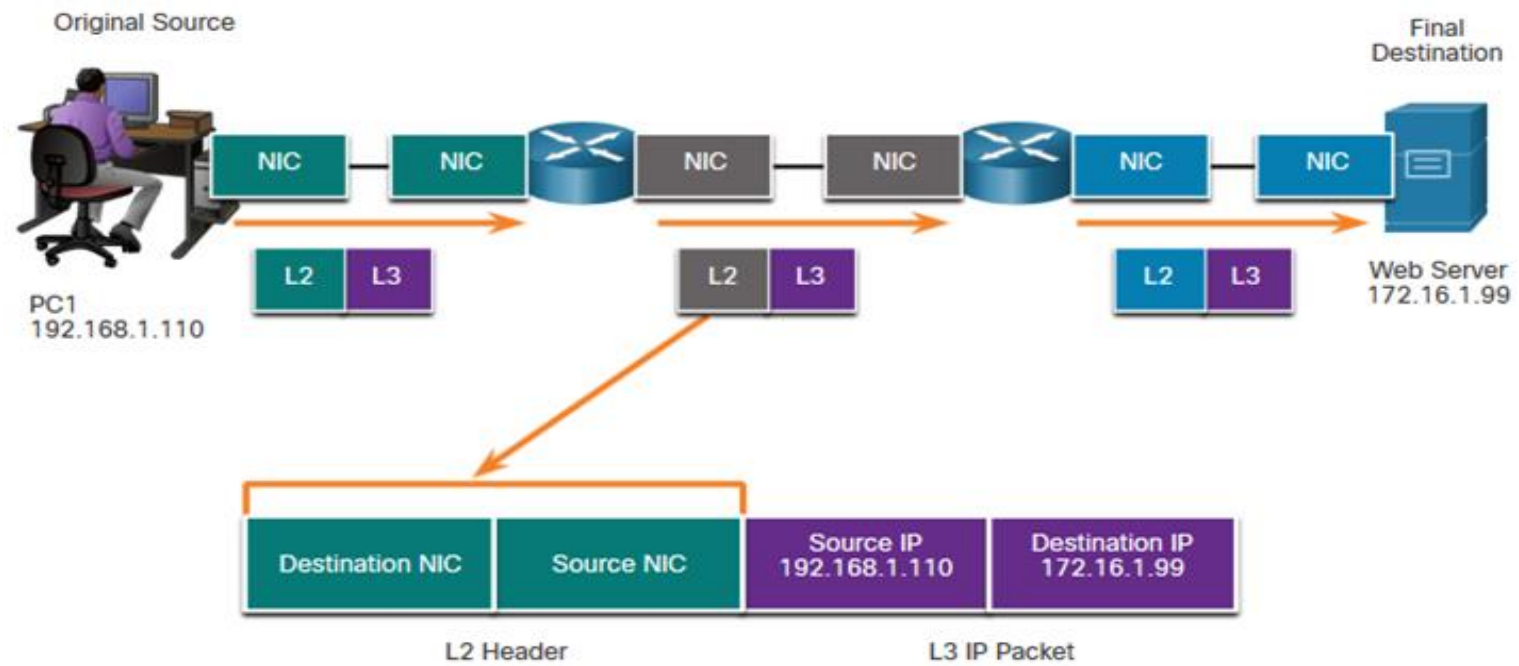
Data Access

Data Link Addresses



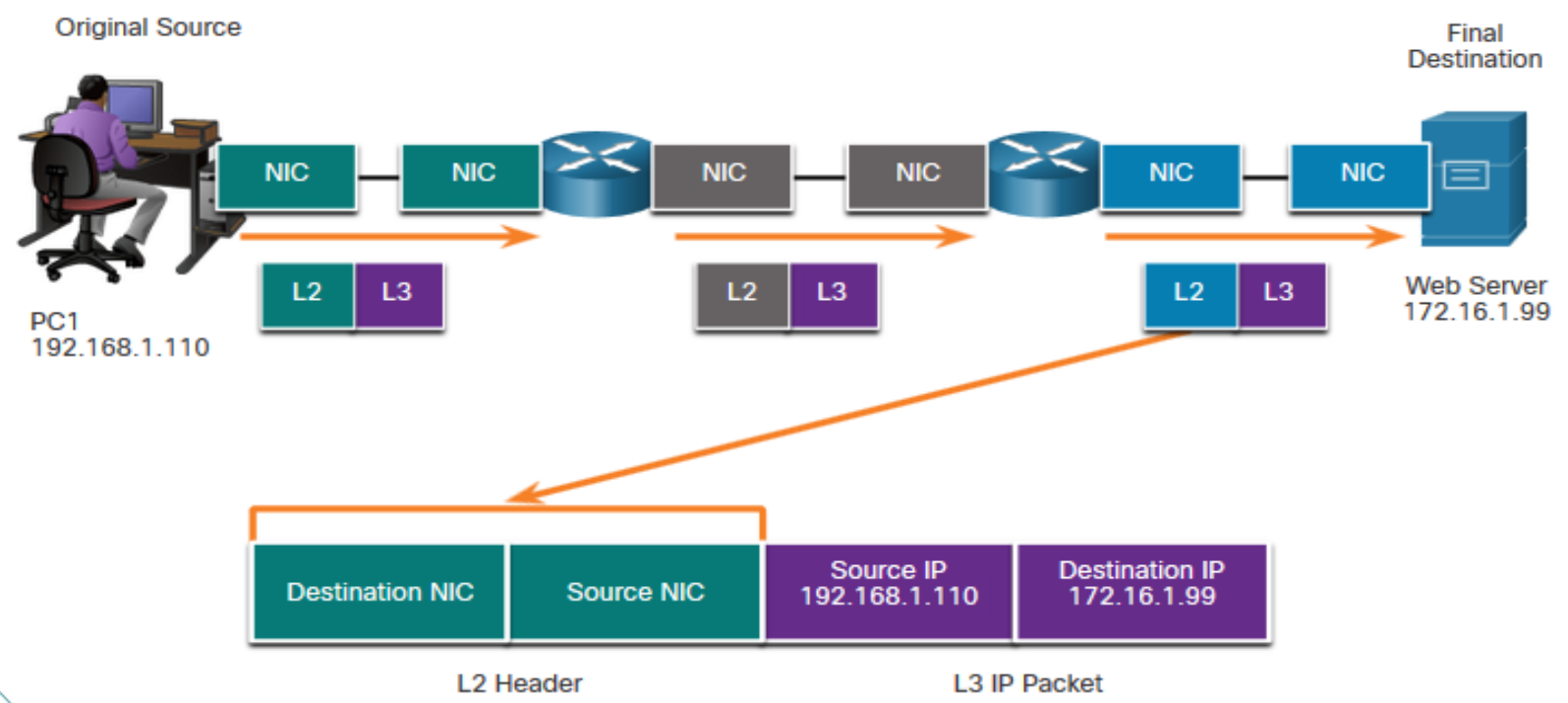
Data Access

Data Link Addresses (Cont.)



Data Access

Data Link Addresses (Cont.)



Summary / Overview

Cheat Sheet

